

Exploring Steganography in 3D Manufacturing Models

Rakesh Kumar¹, Savina Bansal¹, and RK Bansal¹

¹Deptt of ECE, GZS Campus, MRSPTU, Bathinda, Punjab, India
{kumarrakesh0791, savina.bansal, drrakeshkbansal}@gmail.com

Abstract - *With the proliferation of additive manufacturing and 3D computer-aided design (CAD), new opportunities have emerged to leverage steganography techniques in 3D models destined for manufacturing. This paper reviews steganographic techniques for 3D meshes and explore their applications in concealing secret information within manufactured objects and engineering models. Furthermore, key challenges associated with 3D steganography are also discussed. By bridging information hiding techniques with digital manufacturing processes, steganography shows promise for augmenting product security, certification, and quality assurance in ways not feasible with conventional watermarking or overt tagging.*

Keywords - *Steganography, 3D models, CAD.*

1. Introduction

With the proliferation of 3D scanning, modeling, and printing technologies, triangular polygon meshes have become ubiquitous representations for physical objects and engineered designs across manufacturing industries [1]. The flexibility of the 3D mesh structure also provides opportunities for covert communication via information hiding techniques like steganography [2]. By imperceptibly altering 3D models and objects, confidential messages can be embedded and extracted by authorized parties without affecting primary model utility. This has growing relevance in computer-aided manufacturing, intellectual property protection, supply chain tracking, and quality assurance.

Steganography aims to conceal secret messages within cover objects like images, audio, video, or 3D models [1] [3]. The key requirements are high hidden data capacity, minimal perceptual distortion, and security against detection or removal of the hidden data.

This paper reviews advances in steganographic techniques for 3D triangle meshes relevant to manufacturing use cases. Pragmatic applications like hiding manufacturing or design details within engineering models, tracing 3D printed objects back to source files, and embedding production

metadata or quality measures into manufactured components are discussed with suitable examples. Furthermore, open challenges associated with industrial use case are also highlighted.

2. Related Work

Steganography techniques have traditionally focused on hiding messages within 2D image files by imperceptibly altering pixel values or transform domain coefficients [4] [5] [6] [7] [8] [9] [10] [11]. However, with the proliferation of 3D graphics and meshes across industrial and entertainment applications, new opportunities have emerged to leverage 3D models as carriers for hidden communication.

Early research into 3D mesh steganography focused primarily on hiding messages within the inherent geometric properties and spatial relationships of mesh models. Cayre and Macq [12] proposed one of the first practical spatial domain techniques based on treating each triangle edge as a binary state encoder. By subtly shifting vertex projections across edge thresholds, a 1-bit message could be embedded per vertex traversal. This provided synchronization without rearrangement but only 1 bit per vertex in capacity. Wang and Cheng [13] expanded on this two-state approach using multiple geometric tests like projection magnitudes and dihedral angles as modulation states to increase capacity to 3 bits per vertex. However, each test still acted as an independent 1-bit channel. Visually large vertex modifications also risked easier detection. Chao et al. [14] advanced capacity through a multilayer technique that divides axis intervals between two reference vertices into interleaved binary state partitions. By offsetting boundaries in each added layer, more extreme vertex shifts are possible before crossing into the next state region. With 23-bit precision, up to 69 bits per vertex were achievable by stacking layers. But visual distortion and synchronization errors scale with more layers as vertices approach state boundaries. Ohbuchi et al. [15] introduced an entirely different approach using quantization index modulation. Vertices were partitioned into bins based on geometric properties like projections or volume ratios and bin indices selected based on hidden bits. This provided synchronization without rearrangement but remained limited to 2-3 bits per vertex.

Rather than altering spatial geometry, permutation domain methods rearrange vertex order to encode messages in the resulting sequence relative to some canonical ordering [16] [17]. This intrinsically avoids distortion and provides synchronization through shared reference orders. However, optimal capacities reach only $n \log(n)$ bits for n vertices with high complexity.

Neighborhood schemes reduce this complexity by constraining re-orderings in trade-off with lower capacity.

Bit-plane approaches like Yang et al. [18] sidestepped spatial and structural techniques by hiding directly in vertex coordinate bit strings through least significant bit (LSB) replacement. By overwriting LSB layers with up to 23 bits of message per floating point coordinate, nearly 70 bits could be embedded per vertex with minimal perceptual impact. But incremental distortions at higher layers create detectable patterns. Trying to minimize detection, Zhu et al. [19] developed a multivariate Gaussian model for additive distortion steering on 3D meshes.

More recent trends explore machine learning for improved 3D steganalysis [20] [21], featuring trained classifiers that detect artifacts in statistical feature sets extracted from stego meshes. This has driven investigation into more sophisticated [22] adaptive hiding techniques. However, applications in manufacturing and for industrial use cases remain relatively unexplored compared to conventional digital media. The next section will explore the possible applications of 3D steganography techniques in manufacturing industries.

3. Applications in Manufacturing

Concealing secret information in 3D models and components have promising applications in computer-aided manufacturing, product design, and industrial quality control. This section highlights four prospects where steganography can be beneficial.

3.1 Protection of CAD Model Intellectual Property

Unauthorized distribution of proprietary CAD models can lead to counterfeit production and loss of competitive advantage. Steganography provides a covert channel to protect intellectual property. For example, small signature patterns encoding authorship can be hidden in LSB layers of CAD model vertex coordinates. Alternatively, model permutations can conceal signatures. If models are intercepted during transmission or exfiltration, the signatures can trace leaks back to internal sources without overt watermarks.

3.2 Traceability of 3D Printed Objects

3D printing from standard model files produces identical printable objects. This facilitates counterfeiting and inhibits tracing physical instances back to digital source files. By hiding unique instance identifiers in printed objects, steganography provides a way to covertly fingerprint 3D

printed items. For example, multilayer or permutation methods can embed serial numbers during slicing then extracted from the physical build.

3.3 Quality Control Signaling in Manufactured Parts

Steganography presents opportunities to imperceptibly embed quality control and testing metadata in manufactured components. These covert signals have applications in defect tracking, liability, safety recalls, and verifying requirements conformance. For instance, identifiers mapping manufactured parts to quality metrics or inspection results can be embedded in two-state patterns or LSB layers. Parts that fail testing or deviate from specifications are thus surreptitiously tagged. Later extraction of the metadata supports failure tracking, recall identification, or forensic investigation.

3.4 Authentication of CNC Machined Components

Certifying the provenance and conformance of machined metal components is challenging with conventional digital watermarking. But steganographic techniques can subtly alter toolpaths and machine settings to induce microscopic structure permutations, irregularities, and material signatures during machining. These intrinsic fingerprints enable later authentication of genuineness and derivation while being largely invisible to casual inspection. For example, calibrated deviations in cutter positioning, path sequencing, or feed rate can encode identifiers linked to machining instructions and parameters imperceptibly as minor furrows and texture variations. This leaves a fingerprint in the physical part revealing its manufacturing history and conformance to expected instructions.

4. Challenges and Future Outlook

Practical adoption of steganography for manufacturing faces challenges including model distortion, robust extraction, seamless workflow integration, and steganalysis resistance. Most methods induce geometric deviations that accumulate as capacity increases, risking visible artifacts in prints or loss of engineering tolerances. Large payloads also increase probability of detection. Efficient synchronization, error correction, and adaptive hiding will help balance robustness and transparency.

Hiding techniques will need to mesh cleanly with common model and data file formats, slicing software, printers, and scanner acquisition pipelines for transparent integration. Finally, improved

steganalysis resistance is critical if methods are deployed among untrusted parties, calling for robust statistical methods.

With careful technique selection and capacity control, information hiding shows promise for IP protection, traceability, quality assurance, and certification applied to digital manufacturing workflows and physical parts and products. Ongoing steganography research tackling the above issues will help transition techniques into manufacturing practice.

5. Conclusion

This paper reviewed advances in 3D mesh steganography relevant to hiding information within manufactured objects and engineering models. Focus is placed on high-capacity techniques with low perceptual distortion relevant to computer-aided manufacturing. Applications highlighted opportunities in intellectual property protection for CAD models, tracing 3D printed objects to source files, embedding metadata in manufactured parts for quality control, and fingerprinting CNC machined components. Further advancement of capacity, transparency, robustness, and steganalysis resistance will aid adoption within manufacturing industries.

References

- [1] A. Girdhar and V. Kumar, "Comprehensive survey of 3D image steganography techniques," *IET Image Processing*, vol. 12, no. 1, pp. 1–10, Jan. 2018, doi: 10.1049/iet-ipr.2017.0162.
- [2] H. Zhou, K. Chen, Z. Ma, F. Wang, and W. Zhang, *Triangle Mesh Watermarking and Steganography*. Singapore: Springer Nature Singapore, 2023. doi: 10.1007/978-981-19-7720-6.
- [3] I. J. Kadhim, P. Premaratne, P. J. Vial, and B. Halloran, "Comprehensive survey of image steganography: Techniques, Evaluations, and trends in future research," *Neurocomputing*, vol. 335, pp. 299–326, Mar. 2019, doi: 10.1016/j.neucom.2018.06.075.
- [4] S. Kaur, S. Bansal, and R. K. Bansal, "Steganography and classification of image steganography techniques," in *2014 International Conference on Computing for Sustainable Global Development (INDIACom)*, New Delhi, India: IEEE, Mar. 2014, pp. 870–875. doi: 10.1109/IndiaCom.2014.6828087.
- [5] Sumeet Kaur, Savina Bansal, and R. K. Bansal, "An Efficient Adaptive Data Hiding Scheme for Image Steganography," in *Proceedings of the International Congress on Information and Communication Technology*, vol. 438, S. C. Satapathy, Y. C. Bhatt, A. Joshi, and D. K. Mishra, Eds., in *Advances in Intelligent Systems and Computing*, vol. 438, Singapore: Springer Singapore, 2016, pp. 371–379. doi: 10.1007/978-981-10-0767-5_40.
- [6] S. Kaur, S. Bansal, and R. K. Bansal, "Estimating the Effect of Noises over an Efficient Image Steganography Algorithm Based on Intensity and Path Adaptation," vol. 5, no. 1, 2018.
- [7] S. Kaur, S. Bansal, and R. K. Bansal, "Image steganography for securing secret data using hybrid hiding model," *Multimed Tools Appl*, vol. 80, no. 5, pp. 7749–7769, Feb. 2021, doi: 10.1007/s11042-020-09939-7.

- [8] S. Singh, S. Bansal, and S. Singh, "Robust and Secure Image Watermarking using DWT-SVD and Chaotic Map," vol. 4, no. 9, 2015.
- [9] S. Kaur, S. Bansal, and R. K. Bansal, "Reversible Image Steganography Based on Interpolation and Adaptive Approach," *ijcse*, vol. 7, no. 5, pp. 1394–1398, May 2019, doi: 10.26438/ijcse/v7i5.13941398.
- [10] S. Kaur, S. Bansal, and R. K. Bansal, "A Data Security Approach Based on Steganography," vol. 3, no. 2394, 2019.
- [11] S. Bansal, R. K. Bansal, and R. Kumar, "A Novel Upgraded Uniform Embedding Technique for JPEG Steganography," in *Intelligent Sustainable Systems*, vol. 333, A. K. Nagar, D. S. Jat, G. Marín-Raventós, and D. K. Mishra, Eds., in Lecture Notes in Networks and Systems, vol. 333, Singapore: Springer Nature Singapore, 2022, pp. 723–730. doi: 10.1007/978-981-16-6309-3_68.
- [12] F. Cayre and B. Macq, "Data hiding on 3-D triangle meshes," *IEEE Trans. Signal Process.*, vol. 51, no. 4, pp. 939–949, Apr. 2003, doi: 10.1109/TSP.2003.809380.
- [13] C. Wang and Y. Cheng, "An Efficient Information Hiding Algorithm for Polygon Models," *Computer Graphics Forum*, vol. 24, no. 3, pp. 591–600, Sep. 2005, doi: 10.1111/j.1467-8659.2005.00884.x.
- [14] Min-Wen Chao, Chao-hung Lin, Cheng-Wei Yu, and Tong-Yee Lee, "A High Capacity 3D Steganography Algorithm," *IEEE Trans. Visual. Comput. Graphics*, vol. 15, no. 2, pp. 274–284, Mar. 2009, doi: 10.1109/TVCG.2008.94.
- [15] R. Ohbuchi, S. Takahashi, T. Miyazawa, and A. Mukaiyama, "Watermarking 3D Polygonal Meshes in the Mesh Spectral Domain".
- [16] S. Tu, H. Hsu, and W. Tai, "Permutation Steganography for Polygonal Meshes Based on Coding Tree," *IJVR*, vol. 9, no. 4, pp. 55–60, Jan. 2010, doi: 10.20870/IJVR.2010.9.4.2790.
- [17] S.-C. Tu and W.-K. Tai, "A high-capacity data-hiding approach for polygonal meshes using maximum expected level tree," *Computers & Graphics*, vol. 36, no. 6, pp. 767–775, Oct. 2012, doi: 10.1016/j.cag.2012.06.002.
- [18] Ying Yang, N. Peyerimhoff, and I. Ivriissimtzis, "Linear Correlations between Spatial and Normal Noise in Triangle Meshes," *IEEE Trans. Visual. Comput. Graphics*, vol. 19, no. 1, pp. 45–55, Jan. 2013, doi: 10.1109/TVCG.2012.106.
- [19] J. Zhu, Y. Zhang, X. Zhang, and X. Cao, "Gaussian Model for 3D Mesh Steganography," *IEEE Signal Process. Lett.*, vol. 28, pp. 1729–1733, 2021, doi: 10.1109/LSP.2021.3107777.
- [20] Y. Wang, L. Kong, Z. Qian, G. Feng, X. Zhang, and J. Zheng, "Breaking Permutation-Based Mesh Steganography and Security Improvement," *IEEE Access*, vol. 7, pp. 183300–183310, 2019, doi: 10.1109/ACCESS.2019.2960455.
- [21] Z. Li and A. G. Bors, "Steganalysis of meshes based on 3D wavelet multiresolution analysis," *Information Sciences*, vol. 522, pp. 164–179, Jun. 2020, doi: 10.1016/j.ins.2020.02.061.
- [22] I. Yoo *et al.*, "Deep 3D-to-2D Watermarking: Embedding Messages in 3D Meshes and Extracting Them from 2D Renderings." arXiv, Mar. 29, 2022. Last Accessed: Oct. 16, 2023. [Online]. Available: <http://arxiv.org/abs/2104.13450>